MAHATMA GANDHI UNIVERSITY, KOTTAYAM
October 2026 Examination – Fifth Semester

MG5DSEBOTA07 – Plant Protection Strategies: Management of Insect Pest, Weeds and Pathogens

Maximum Marks: 70

PART A – VERY SHORT ANSWER TYPE

Answer any 10 questions (2 marks each)

1. Mention one limitation of pheromone-based methods.

Pheromones are species-specific. They may not provide immediate control of all pest stages.

2. What is conservation of natural enemies?

Protection and maintenance of beneficial organisms such as predators, parasitoids and microbial pathogens.These organisms naturally suppress pest populations.

3. What is a biennial weed?

A weed that completes its life cycle in two growing seasons. Vegetative growth occurs in the first year.Flowering and seed production occurs in the second year.

4. Mention one way to reduce crop–weed competition.

Timely weeding, mulching, crop rotation, optimum crop spacing. (any one)

5. What is plant pathogenic nematodes?

Microscopic, worm-like organisms that attack plants. They may damage roots or other plant parts and reduce plant growth.

6. Why is selectivity important in crop production?

Selectivity allows control of target pests or weeds.It minimizes damage to crops and beneficial organisms.

7. Mention one reason for studying plant pathogenic organisms.

To identify the cause of plant diseases.

To develop suitable prevention and control measures. (any one)

8. Mention one purpose of a field weed survey.

To identify weed species present in a field.

To estimate their abundance and distribution and plan control measures.

9. Name the causative agent of leaf spot disease in Amaranthus.

The causative agent of leaf spot disease in Amaranthus is commonly Cercospora amaranthi.

10. What is targeted pesticide application?

Application of pesticide to a specific target pest, infested plant or affected area. It reduces unnecessary pesticide use and exposure.

11. What is herbicide dosage?

The quantity of herbicide applied per unit area, or the prescribed concentration used for weed control.

12. Mention one biological method of pest management.

Use of natural enemies such as predators or parasitoids.

Acceptable example: use of a suitable microbial biopesticide.

PART B – SHORT ANSWER TYPE

Answer any 8 questions | 5 marks each

13. Describe the role of plant quarantine in protecting pest-free areas.

Prevents introduction of pests and pathogens into pest-free areas. Regulates movement of seeds, plants and planting materials. Uses inspection, testing and certification. May require treatment or destruction of infected material. Protects agriculture, local ecosystems and biodiversity.

14. Explain the effect of insect pests on crop productivity.

Insect pests damage leaves, stems, roots, flowers, fruits and seeds.

Cause injury by chewing plant tissues or sucking sap.

Some insect pests transmit plant pathogens, including viruses.

Reduce crop growth, yield and product quality.

Increase production costs and post-harvest losses.

15. Explain the role of weed monitoring in integrated weed management.

Identifies weed species and estimates their abundance.

Determines weed distribution and level of infestation.

Helps to select appropriate cultural, mechanical, biological or chemical controls.

Enable timely control before flowering and seed production of weeds.

Helps to assess control effectiveness and detect possible herbicide resistance.

16. Explain herbicide formulations.

Herbicide formulation combines the active herbicide ingredient with other materials to aid storage, handling and application.

EC (Emulsifiable Concentrate): Liquid active ingredient dissolved in petroleum solvent with an emulsifier. Forms a milky emulsion in water.

WP (Wettable Powder): Dry powder suspended in water with wetting agents. Requires continuous agitation in the spray tank.

G / GR (Granules): Active ingredient applied onto inert granules (e.g., clay, silica). Applied directly to soil without water.

SL (Soluble Liquid): Liquid active ingredient that completely dissolves in water to form a clear solution.

SC / FS (Suspension Concentrate): Solid active ingredient finely ground and suspended in a liquid base.

17. Explain the relationship between causative agent, symptoms and control. (Liberal valuation)

Causative agent: organism or factor responsible for a disease, e.g. fungus, bacterium, virus or nutrient deficiency.

Symptoms: visible changes such as leaf spots, wilting, yellowing, rotting or stunting.

Diagnosis links symptoms with the likely cause and may require laboratory tests.

Control is chosen according to the cause: sanitation, resistant varieties, cultural practices or suitable pesticides.Correct diagnosis is essential because different causes can produce similar symptoms.

18. Write a short essay on banana bunchy top disease.

Causal agent: Banana bunchy top virus (BBTV).

Vector: banana aphid, Pentalonia nigronervosa.

Symptoms: dark-green streaks, narrow upright leaves, shortened internodes and a bunchy appearance at the top.

Spread: infected planting material and viruliferous aphids.

Management: use disease-free planting material, remove and destroy infected plants, manage the vector and follow quarantine measures.

19. Explain the significance of resistant host plants.

Resistant varieties reduce pest and disease damage.

Reduce dependence on chemical pesticides.

Can lower production costs and environmental risks.

Provide useful protection as part of integrated pest management.

Resistance may involve antixenosis (non-preference), antibiosis (adverse effect on pest) or tolerance (withstanding damage).

20. Explain diseases due to abiotic causes.

Abiotic disorders arise from non-living environmental or chemical factors rather than infectious organisms.

Nutrient deficiencies may cause chlorosis and poor growth.

Water stress causes wilting; temperature extremes cause heat or chilling injury.

Salinity, unsuitable soil pH, poor aeration and chemical injury may damage plants.

Management: correct nutrient deficiencies, ensure proper irrigation and drainage, improve soil conditions and avoid chemical misuse.

21. Explain Integrated Pest Management (IPM). Mention the important points involved.

IPM combines compatible methods to keep pests below economically damaging levels while minimizing risks to humans and the environment.

Correctly identify pests and natural enemies.

Monitor pest populations and use economic/action thresholds where applicable.

Use cultural, mechanical, physical and biological methods, and resistant varieties.

Apply selective pesticides judiciously when needed; evaluate results and environmental safety.

22. Explain IPM strategies with suitable examples.

Cultural: crop rotation, field sanitation and timely sowing.

Mechanical: hand-picking pests and removing infested plant parts.

Physical: traps, barriers and suitable heat treatments.

Biological: Trichogramma parasitoids or Bacillus thuringiensis against susceptible caterpillars.

Host resistance and need-based use of suitable selective pesticides, supported by regular monitoring.

PART C – ESSAY TYPE

Answer any 1 question | 10 marks

23. Discuss the weed classification and various methods of weed control.

Weeds are unwanted plants growing where they are not desired and competing with cultivated plants.

Classification by life cycle:- annual, biennial and perennial weeds.

Classification by morphology:-

Broadleaf Weeds (Dicots): Expanded net-veined leaves, taproot system. Examples: Parthenium hysterophorus, Convolvulus arvensis.

Grasses (Monocots): Narrow leaves with parallel veins, hollow stems with nodes, fibrous roots. Examples: Echinochloa, Eleusine indica.

Sedges (Monocots): Solid, triangular stems without nodes; 3-ranked narrow leaves. Example: Cyperus rotundus, Cyperus esculentus.

Classification by habitat:

Terrestrial: Grown on land (dryland or irrigated fields).

Aquatic:

1.Submerged: Hydrilla verticillata, Vallisneria.

2.Floating: Eichhornia crassipes (Water hyacinth),Salvinia molesta.

3.Emergent: Typha latifolia.

Special Ecological Categories - Parasitic Weeds:

Total Root Parasite: Orobanche spp. (Broomrape) on Mustard/Solanaceous crops.

Partial Root Parasite: Striga spp. (Witchweed) on Sorghum/Sugarcane.

Total Stem Parasite: Cuscuta spp. (Dodder)

Partial Stem Parasite: Dendrophthoe falcata (Loranthus) on fruit trees

Control measures:-

Cultural control: crop rotation, mulching, proper spacing and cover cropping.

Mechanical control: hand weeding, hoeing, tillage and mowing.

Physical control: heat treatment, solarization and barriers.

Biological control: use of natural enemies or pathogens to suppress weeds.

Chemical control: suitable selective or non-selective herbicides, applied according to label directions.

Integrated weed management combines suitable methods to reduce crop losses and environmental harm.

24. Define insect pest management. Write an essay on insect pests of cowpea and their management.

Insect pest management is the identification, monitoring and control of insect pests to minimize crop damage and economic losses.

Insect pests of Cow pea

1.Gram pod borer: Helicoverpa armigera

Symptoms:

Defoliation in early stages

Larva’s head alone thrust inside the pods and the rest of the body hanging out.

Pods with round holes

Management:

Collection and release of potential natural enemies viz., grubs and adults of (Lady bird beetles) Coccinella transversalis, Cheilo menes sexmaculatus, Harmonia octo maculata and maggots of Ischiodon scutellare.

Spraying of neem oil emulsion 3 % or Dimethoate 2 ml/l at the time of aphid incidence.

2. Spotted pod borer: Maruca testulalis

Symptoms:

Bore holes on the buds, flower or pods

Infested pods and flowers are webbed together.

Management:

Avoid closer plant spacing.

Collect and destroy larvae and adults to the extent possible

Install pheromone traps at a distance of 50 m

Collection and release of potential natural enemies.